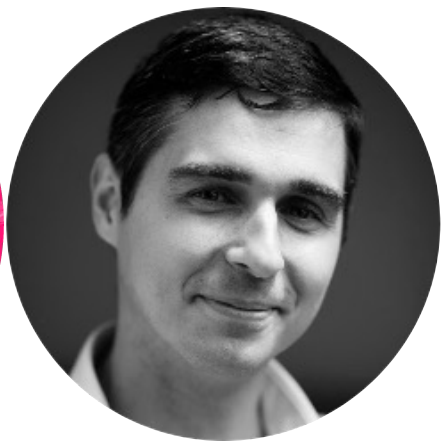


Java SE 21 Developer (Exam 1Z0-830): Controlling Program Flow

Understanding Basic Conditional Logic



Andrejs Doronins

Software Developer in Test



Why Won't This Compile?

```
int day = Integer.parseInt(args[0]);

String result = switch (day) {
    case 1 -> "Monday";
    case "2" -> "Tuesday";
    case 6,7 -> { yield "Weekend"; };
};

System.out.println(result);
```



Be the compiler...



Course Introduction

Basic conditional logic

- if-else, ternary expressions, switch statements

Switch expressions

- More powerful (and complex) than switch statements

Pattern matching

- Safer and shorter type casting
- instanceof

Looping

Advanced control flow (in loops)



must be a boolean

```
if (condition) {  
    // branch if true  
}
```

curly braces are NOT
(always) mandatory



must be a boolean

```
if (condition) {  
    // statement 1  
    // statement 2  
}
```

curly braces ARE required
for 2+ statements



JavaScript Code Example

won't work in Java



```
if(1) {  
    console.log("1 is truthy");  
}  
if(0) {  
    console.log("...")  
}
```



```
// assignment, not comparison  
if(x = 1) { // doesn't compile  
  
}
```

```
// OK  
if(x == 1) {  
  
}
```



What Will These Print?

```
if(false) {  
    System.out.println("inside");  
    System.out.println("inside?");  
}
```

```
if(false)  
    System.out.println("inside");  
    System.out.println("inside?");
```

This one line will print



must be a boolean



```
if (condition) {  
    // branch if true  
} else {  
    // branch if false  
}
```



```
if (condition1) {  
    // branch if true  
} else if (condition2) {  
    // branch if true  
} else {  
  
}
```



Ternary Expression

true

booleanExpression ? expression1 : expression2

false



The diagram illustrates the execution of a ternary expression. A purple arrow labeled 'true' points from the 'booleanExpression' part to 'expression1'. A pink arrow labeled 'false' points from the 'booleanExpression' part to 'expression2'.

`int x = 5 > 4 ? 1 : 2;`

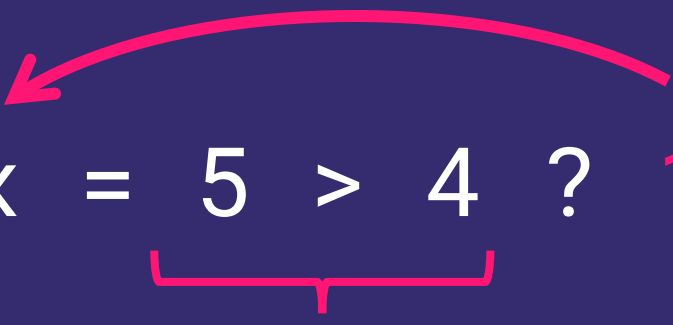


Diagram illustrating the evaluation of the ternary operator `5 > 4 ? 1 : 2`. The expression `5 > 4` is evaluated as `true`, and the result `1` is assigned to `x`.

`int x = 5 < 4 ? 1 : 2;`

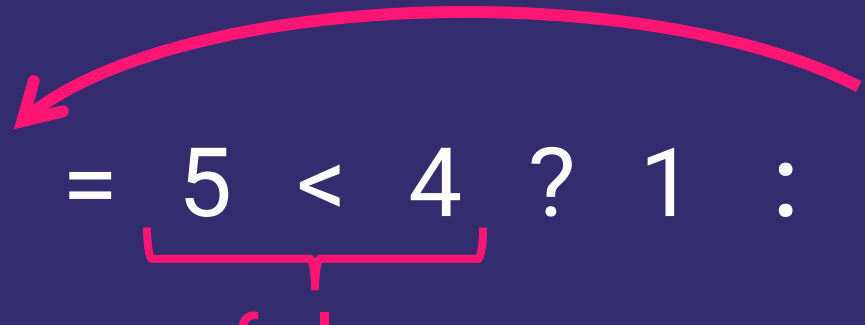


Diagram illustrating the evaluation of the ternary operator `5 < 4 ? 1 : 2`. The expression `5 < 4` is evaluated as `false`, and the result `2` is assigned to `x`.



```
if(condition) {  
    // return a value  
    // or run some code  
} else {  
    // same here  
}
```

```
var x = boolExpr ? x : y ;
```

```
var y = boolExpr ? 5 : "abc";
```



does not compile



Question 1

```
int x = 5;  
if(x < 5)  
    System.out.println("a");  
    x++;  
System.out.println("b");  
System.out.println(x);
```

b
6



Question 2

```
int y = 2;  
if(y = 2) {  
    System.out.println("a");  
    y++;  
} else {  
    System.out.println("b");  
    System.out.println("c");  
}  
System.out.println(y);
```

compilation
failure



Question 3

```
int x = 1;  
int y = 2;  
int z = x > y ? 2 : 3.0;
```

```
System.out.println(z);
```

compilation
failure



boolean

```
if(condition) {  
    // code  
} else {  
    // code  
}
```

byte Byte
short Short
int Integer
char Character

```
switch (var) {  
    case option1:  
        // code  
    case option2:  
        // code  
}
```

String
enum
(Since Java 21)
All Object types



```
if(condition) {  
    // code  
} else {  
    // code  
}
```

optional



```
switch (var) {
```

```
    case option1:  
        // code  
    case option2:  
        // code
```

```
}
```

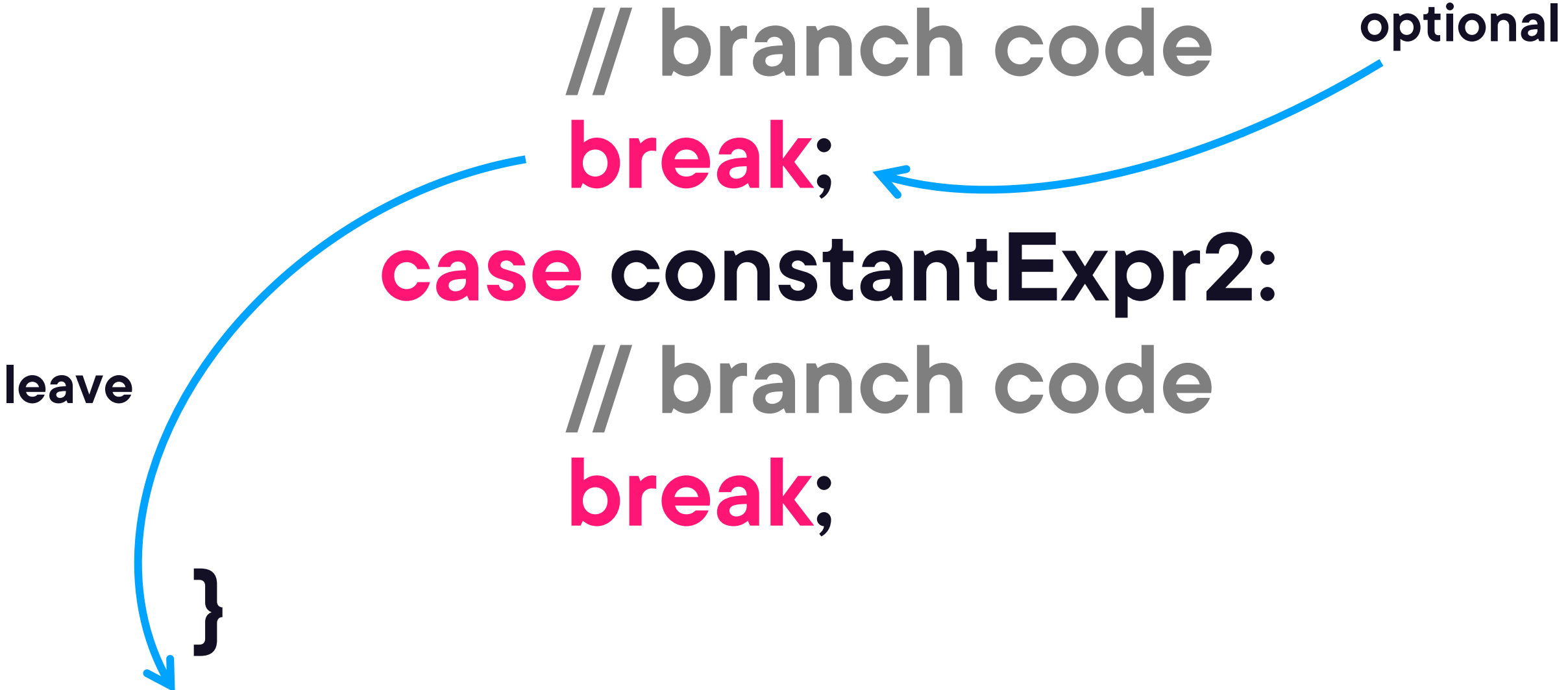
required



```
switch (testVar) {  
    case constantExpr1:  
        // branch code  
        break;  
    case constantExpr2:  
        // branch code  
        break;  
}
```

optional

leave



```
Month month = Month.JUNE;
```

```
switch (month) {  
    case DECEMBER:  
        System.out.println("Winter");  
        break;  
    case APRIL:  
        System.out.println("Spring");  
        break;  
    case JUNE:  
        System.out.println("Summer");  
        break;  
}
```



```
Month month = Month.JUNE;
```

```
switch (month) {  
    case DECEMBER:  
        System.out.println("Winter");  
        break;  
    case APRIL:  
        System.out.println("Spring");  
        break;  
    default:  
        System.out.println("oops");  
        break;  
}
```



```
Month month = Month.JUNE;
```

```
switch (month) {  
    case DECEMBER:  
        System.out.println("Winter");  
        break;  
    default:  
        System.out.println("oops");  
        break;  
    case JUNE:  
        System.out.println("Summer");  
        break;  
}
```



**Branch order
doesn't matter**



```
switch (testVar) {  
    case constantExpr1:  
        // branch code  
        break;  
    case constantExpr2:  
        // branch code  
        break;  
    default:  
        // branch code  
        break;  
}
```



```
Month month = Month.JUNE;
```

```
switch (month) {  
    case JUNE:  
        System.out.println("Summer");  
        break;  
    case DECEMBER:  
        System.out.println("Winter");  
        break;  
    case APRIL:  
        System.out.println("Spring");  
        break;  
}
```



Output

Summer
Winter
Spring



```
Month month = Month.JUNE;
```

```
switch (month) {  
    case DECEMBER:  
        System.out.println("Winter");  
    case JUNE:  
        System.out.println("Summer");  
    case APRIL:  
        System.out.println("Spring");  
}
```



Output

Summer
Spring



```
switch (month) {  
    case DECEMBER:  
    case JANUARY:  
    case FEBRUARY:  
        System.out.println("Winter");  
        break;  
    case MARCH:  
    case APRIL:  
        System.out.println("Spring");  
        break;  
    // etc.  
}
```



**Duplicate values
are not allowed**



**The values in each case
clause must be constant and
of the same data type as the
switch value**



```
void doThing(String arg) {  
    final String finalVar = "b";  
    switch (arg) {  
        case "Literal":  
            // ...  
        case finalVar:  
            //...  
    }  
}  
// or "case Month.DECEMBER"
```



```
void doThing(String arg, final String finalArg) {  
  
    String localVar = "a";  
  
    switch (arg) {  
  
        case localVar:   
            // ...  
  
        case finalArg:   
  
            // ...  
        case 5:   
            // ...  
    }  
}
```



Question 1

What is the output of this code?

```
String s = "hi";  
switch (s) {  
    case "HI":  
    case "Hello":  
        System.out.println("a");  
        break;  
    default:  
        System.out.println("b");  
        break;  
}
```

b



Question 2

What is the output of this code?

```
int i = 5;  
switch (i) {  
    case 2:  
        System.out.println("a");  
    case 5:  
        System.out.println("b");  
    default:  
        System.out.println("c");  
        break;  
}
```

b
c



Question 3

What is the output of this code?

```
void runCode(String arg) {  
    int x = 5; String varOne = "abc";  
    switch (arg) {  
        case "Test":  
            System.out.println("a");  
            break;  
        case varOne:  
            x++;  
            break;  
    }  
    System.out.println(x);  
}
```

compilation
failure

